Cellular events during scar-free skin regeneration in the spiny mouse, *Acomys*

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ABSTRACT

In contrast to the lab mouse, *Mus musculus*, several species of spiny mouse, *Acomys*, can regenerate epidermis, dermis, hairs, sebaceous glands with smooth muscle erector pili muscles and skeletal muscle of the panniculus carnonsus after full thickness skin wounding. Here, we have compared the responses of these scarring and nonscarring organisms concentrating on the immune cells and wound cytokines, cell proliferation, and the collagenous components of the wound bed and scar. The blood of *Acomys* is very neutropenic but there are greater numbers of mast cells in the *Acomys* wound than the *Mus* wound. Most importantly there are no F4/80 macrophages in the *Acomys* wound and many proinflammatory cytokines are either absent or in very low levels which we suggest may be primarily responsible for the excellent regenerative properties of the skin of this species. There is little difference in cell proliferation in the two species either in the epidermis or mesenchymal tissues but the cell density and matrix composition of the wound is very different. In *Mus* there are 8 collagens which are up-regulated at least 5-fold in the wound creating a strongly trichrome-positive matrix whereas in *Acomys* there are very few collagens present and the matrix shows only light trichrome staining. The major component of the *Mus* matrix is collagen XII which is up-regulated between 10 and 30-fold after wounding. These results suggest that in the *Acomys* wound the absence of many cytokines resulting in the lack of macrophages is responsible for the failure to up-regulate fibrotic collagens, a situation which permits a regenerative response within the skin rather than the generation of a scar.

The cellular and molecular events of wound repair following damage to the adult mammalian skin has been extensively studied and is commonly described in terms of three phases.[1] First, there is an inflammatory phase where following blood coagulation by platelets, neutrophils, and then monocytes enter the wound from the severed capillaries to phagocytose contaminating bacteria and tissue debris and release further chemotactic and growth factors. In the second phase granulation tissue is created consisting of macrophages, proliferating fibroblasts, and neovascularization surrounded by a loose matrix of fibronectin, collagens, and hyaluronic acid. This granulation tissue is covered externally by a new wound epidermis which has migrated from the cut edges through or underneath the scab to reestablish epithelial continuity. In the third phase, the provisional matrix of the granulation tissue is remodeled with the elimination of molecules such as fibronectin and the accumulation of others such as collagen type I. This sequence of events results in a smooth, hairless epidermis covering a dermis which has failed to replace any of those structures removed such as the neatly layered dermis or hairs and their associated structures and produces a uniform scar consisting of thick bundles of collagen running parallel to the epidermis.

In contrast, adult fish and Urodele skin regenerates perfectly after wounding,[2] the former regenerating bony scales and the latter, including terrestrial species regenerating either a mucogenic or a keratinized epidermis, dermis, and dermal glands.[3,4] It would be simple to assume that this regenerative property has been lost in mammals, but that may not be the case. Holes punched through rabbit ears,[5] the murphy roths large (MRL) strain of mice,[6] and spiny mice of the genus *Acomys*[7] regenerate the skin and hairs during the process of tissue replacement and wounds made to the skin of mammalian fetuses up to the middle of the third trimester of gestation also heal perfectly.[8–11] Nevertheless, it is certainly the case that perfect mammalian skin regeneration is not universal as it is in lower vertebrates.

To identify potential reasons for this loss/reduction in skin regenerative ability in adult mammals the most productive comparisons have been made between mouse embryos and adults.[8–11] Fetal wounds show a blunted inflammatory response: embryonic platelets release less PDGF, TGFβ1, and TGFβ2,[12] there are fewer neutrophils in the embryonic wound,[13] there are fewer macrophages in the nonscarring stage embryo and they are less persistent[14] and there are virtually no macrophages at all in the early embryo.[15] All of this results in greatly reduced levels of inflammatory cytokines such as IL-6,[16] IL-8,[17] and growth factors such as TGFβs.[18] Mast cells are other immune cells thought to play a role in wound healing by releasing cytokines to attract neutrophils to the wound. In their absence neutrophil recruitment is lower[19] and as their numbers are far lower in fetal scarless wounds they are considered to augment scarring in the adult.[20] Similarly, the lower vertebrate immune system is underdeveloped relative to mammals with, e.g., fewer neutrophils[3] and a reduced cytokine response following wounding.[21] In Anurans the ability to regenerate whole limbs disappears after metamorphosis coincident with the development of an immune system producing an inflammatory response in injured tissues[22] and *Rana* skin transitions from scar-free healing to scarring over this same period.[23]

Another difference between fetal and adult skin in their response to wounding lies in the composition of the matrix. The collagen III: collagen I ratio is higher in fetal skin[8] and these fibers regenerate in a reticular pattern in the wound rather than a parallel fiber, highly cross-linked arrangement seen in the adult wound.[24] There are higher levels of hyaluronic acid, fibronectin, and chondroitin sulphate levels in fetal skin and wounds[25–27] and the balance between the enzymes responsible for remodeling the extracellular matrix (ECM), the matrix metalloproteases (MMPs) and their inhibitors, the tissue inhibitors of metalloproteases (TIMPs), favors the MMPs in fetal wounds and TIMPs in adult wounds.[28] Again the scar-free regenerating skin of Urodeles shows embryonic features of high MMP levels, a high collagen III: collagen I ratio and high tenascin-C levels.[3]

With these emerging principles in mind we have compared, here, the regenerative response of the adult spiny mouse *Acomys cahirinus* skin to 8-mm full thickness skin wounds with that of the scarring lab mouse, *Mus musculus*. The goal was to specifically examine their immune systems and the cytokine response to wounding, cell proliferation in the wounds, the composition of the wound matrix and MMP and TIMP levels, and the vascularity of the wound bed.

MATERIALS AND METHODS

Animals

Acomys cahirinus were obtained from a breeding colony housed at University of Florida and *Mus musculus* of the CD-1 outbred strain were purchased from Charles River. All experiments were conducted on groups of adult animals of mixed sexes of 6 months to 1 year of age. The experiments were conducted in accordance with the Institutional Review Board on Animal Care at the University of Florida. Mice were anaesthetized with iso-fluorane, the hair shaved and two 8-mm biopsy punch wounds made through the mid-dorsal full thickness skin. At various times after wounding the animals were sacrificed and the wound tissue dissected out including a surrounding ring of normal skin for histology, but excluding surrounding normal skin for microarrays and proteomics. For each experiment and each experimental time point a group size of three to five animals was used.

Histology

For general histological and immunological studies group sizes were $n = 5$ and both wounds of the same animal were harvested. For blood cell counts $n = 3$ unwounded animals of each species were used. For proliferation studies $n = 3$ animals were used. For the blood cell counts animals had their tail tips amputated to generate a drop of blood which was smeared on a glass microscope slide and stained with Geimsa. One thousand cells were counted for each slide and identified according to morphology. For general histology both wounds at each time point were fixed in 4% paraformaldehyde overnight, embedded in paraffin wax and sectioned at 10μm. Masson trichrome staining was performed with a kit purchased from Fisher Scientific. To examine mast cells toluidine blue staining was performed on paraffin wax sections. For immunocytochemistry on paraffin wax sections the dehydrated sections were microwaved for 4 minutes in citrate buffer pH6 for antigen retrieval and the Vectastain Eite ABC kit (Vector Labs) was used with diaminobenzidine as the chromogen and the stained sections dehydrated and mounted in Permount. For immunocytochemistry on frozen sections, fixed tissue was placed overnight in 30% sucrose, embedded in optimal cutting temperature (OTC) and cryosectioned at 20μm. Sections were blocked for 1 hour in 0.5% Triton X-100, 0.1% fish skin gelatin (Sigma), 0.1% Tween, 10% donkey serum in phosphate buffered saline (PBS), then incubated with primary antibodies at a 1 in 100 dilution in blocking solution overnight at 4°C. The following day sections were washed in PBS and incubated for 1 hour in the dark with secondary antibodies at a dilution of 1 in 200 and mounted in 80% glycerol containing 1 in 10,000 Hoeschst. The primary antibodies used in this study were smooth muscle actin (SMA) (rabbit polyclonal, Abcam), antimast cell tryptase antibody (rabbit monoclonal, Abcam), F4/80 (rat monoclonal, Abcam) and collagen XII (rabbit polyclonal, kind gift of Dr D. Birk, University of South Florida and Dr M. Mark, University of Cologne, Germany), and collagen XII (rabbit polyclonal, Sant Cruz). The secondary antibodies used for fluorescence were Alexa Fluor 488 donkey anti-rabbit IgG (Abcam). For proliferation analysis of wounds a Click-iT EDU kit (Life Technologies) was used according to the manufacturer's protocol. An i.p. injection of 100 mg/kg 5-ethynyl uridine (EDU) in PBS was administered to each animal 2 days and 1 day before sacrifice, three animals at each time point had their wounds fixed in 4% para-formaldehyde. 20 μm frozen sections were cut for the Click-iT protocol and sections mounted in 80% glycerol containing 1 : 10,000 Hoescht. Under immunofluorescence EDU+ve cells and total cell numbers (4′,6-diamidine-2-phenylidone (DAPI) positive) were counted in both the epidermis and mesenchyme using five counts (500–1,000 cells) on each sample at each time point were performed.

Cytokine analysis

Cytokine assays were performed on group sizes of $n = 3$ for each species and both wounds on the same animal were pooled. Sample times were day 0 (unwounded skin), day 3, day 5, day 7, and day14 after wounding with 8-mm biopsy punches. The wound samples (100 mg) were cleanly dissected from the animal and homogenized. Soluble proteins were separated by centrifugation at 50,000 × g for 30 minutes at 4°C. Protein levels were measured using a bicinchoninic acid (BCA) protein assay kit (Pierce) and equivalent amounts of protein was profiled using commercially available sandwich enzyme-linked immunosorbent assay according to the manufacturer's instruction (Proteome Profiler Array ARY006; R&D Systems). The average pixel density from each pair of duplicate spots representing each cytokine was determined by subtracting an averaged background pixel density from each spot pixel density using Alphaview version 2 from FluorChem E (Proteinsimple CA). A series of comparison arrays were performed on corresponding signals to determine the relative change in cytokine levels between samples.

Microarrays

For microarray analysis six biological replicates for *Acomys and Mus* were used from sample times of day 0 (unwounded skin), day 7, and day 14 after wounding with 8-mm biopsy punches. For reverse transcription-polymerase chain reaction (RT-PCR), six biological replicates were sampled on days 0, 3, and 5. Total RNA was extracted from RNALater (Qiagen) treated tissue samples using RNeasy Fibrous Tissue Mini Kit (Qiagen Cat. 74704) following the manufactures recommended protocol. RNA concentration and yield was determined using the NanoDrop 1000 instrument (Thermo Scientific, Wilmington, DE). RNA quality was assessed on an Agilent 2100 BioAnalyzer (Agilent, Andover, MA) using an RNA PicoChip. All specimens had RNA Integrity Score (RIN) scores greater than 5. 100 ng of total RNA was processed for microarray analysis using the GeneChip WT PLUS Reagent Kit (Affymetrix, Santa Clara, CA) following manufacturer's recommended protocols. cDNA (5.5 ug) was fragmented, terminally labeled and hybridized to Affymetrix GeneChip Mouse Gene 2.0 ST Array (MoGene 2.0 ST) for 16 hours at 45 °C and washed following Affymetrix's fluidics protocols FS450_001. Microarrays were normalized using the robust multiarray average method as implemented in Partek Genomics Suite 6.6 (Partek Incorporated, St Louis, MO). Only annotated probe sets were used in the subsequent analysis. BRB-ArrayTools (version 4.3.0 Stable Release, Richard Simon & BRB-ArrayTools Development Team [http://linus.nci.nih.gov/BRB-ArrayTools.html]) was utilized to identify significant genes ($p < 0.001$) using the class prediction tool. Leave-one-out-cross-validation studies and Monte Carlo simulations were performed using BRB-Array Tools.

For RT-PCR, cDNA was generated from 500 ng of RNA using iScript cDNA Synthesis Kit (Bio-Rad cat. no. 170-8891) following manufacturer's protocol. Real-time PCR was performed using SsoFast EvaGreen Supermix (Bio-Rad cat. no. 172-5200) following manufacturer's protocol on a Bio-Rad CFX-96 Real-Time PCR Detection System. Fold change of expression was calculated using the ΔΔCt relative expression method. Change in expression was expressed as normal skin vs. wounded skin.

Proteomics

Proteomic analyses were performed on group sizes of $n = 3$ for each species and both wounds on the same animal were pooled. Sample times were day 0 (unwounded skin), day 7, and day14 after wounding with 8-mm biopsy punches. All the experimental procedures were followed as previously described.[29,30] Enzymatically digested proteins were analyzed following 1D PAGE separation. Nanoscale liquid chromatography tandem mass spectrometry (nano-LC-MS/MS) experiments were performed using the Agilent 1100 system connected to a LTQ-Velos mass spectrometer (Thermo Finnigan) equipped with a nano-electrospray ion source. Samples (10 μL) were loaded onto a reversed phase C18 column (Aqua; particle size 5 μm, 100 μm × 15 cm) and separated at a flow rate of 200 nL/minute using a linear 150 minutes gradient from 0 to 50% mobile phase A (mobile phase A: 100% H_20, 0.1% formic acid, mobile phase B: ACN, 0.1% formic acid). LC/MS/MS spectra were acquired in a data-dependent mode. Data were searched with Mascot Daemon (V2.4, Matrix Science, London, United Kingdom) using Swiss plot mouse database (November, 2014).

Statistics

Unpaired *t*-tests with equal variance were performed on the graphical data and the error bars on the graphs represent standard deviations.

RESULTS

General characteristics of skin response to wounding

In the studies reported here we have used 8-mm circular wounds made through full thickness skin of the mid-dorsal region and compared the outbred CD-1 *Mus musculus* strain with *Acomys cahirinus* from our in-house colony. Animals were 6–12 months of age, groups ($n = 3$–5) consisted of mixed sexes and samples were collected after varying time periods, usually every other day, for 35 days. We have previously described a faster rate of reepithelialization of the *Acomys* wound than in *Mus*[7] and this was also the case here. Scab formation and its shedding was faster in *Acomys*, e.g., Figure 1A shows a day 14 *Mus* wound which still has a scab present in contrast to all the *Acomys* wounds which had completed this process several days previously (Figure 1D).

After the completion of scab formation, wound healing and generation of a granulation tissue the histological appearances of the wounds were very different. By day 10, the *Mus* wound had a thick wound epithelium often with rete ridges penetrating the mesenchyme (Figure 1G). The mesenchyme itself was densely packed with cells showing an upward directionality to the matrix and vasculature giving the impression that the granulation tissue had arisen by cell migration from the deeper layers of connective tissue below the muscle of the panniculus carnosus (Figure 1G, black arrows). In contrast, an equivalently staged *Acomys* wound (Figure 1H) had a much looser network of vasculature and cells with the matrix creating horizontal strands joining the tissues layers from side to side rather than the vertical appearance of the *Mus* wound bed.

During the third week the granulation tissue of the *Mus* wound became remodeled as the red staining trichrome tissue was replaced by a collagenous blue staining tissue and matrix (Figure 1I). The wound epithelium above it now approached the thickness of normal epithelium and had differentiated into layers with the stratum corneum on the outside. By the end of the third week the wound bed was a uniform mass of tissue with no cellular structures present apart from the vasculature and it was developing into a collagenous scar (Figure 1K). By contrast the *Acomys* wound bed showed far less trichrome staining, a loosely packed arrangement of cells which had changed little from earlier sampling times and a wound epithelium that had started to generate many hair follicles (Figure 1J). We have previously reported that these regenerated hair follicles reexpress embryonic cytokeratins, Wnt and BMP pathway components[7] and by the end of the third week the *Acomys* wound bed was deeply penetrated with new hair follicles which had started to develop sebaceous glands while the rest of the mesenchyme showed little change (Figure 1L). During the fourth week these newly developed hairs became externally visible in the center of the *Acomys* wound as golden colored spiny hairs (Figure 1F, red arrowheads). During the fifth week the skeletal muscle of the panniculus carnosus had reformed across the *Acomys* wound and fully developed hair follicles had produced erector pili smooth muscles (Figure 1M, stained for SMA, black arrowheads), sebaceous glands (sg), and adipose cells (ad) beginning to repopulate the newly regenerated skin below the hair follicles (Figure 1N).

Proliferative response to wounding

We next considered whether these completely different responses to wounding could be attributable to differing proliferation responses either in the epidermis or mesenchyme. To analyze this, we used EDU injected into animals 48 and 24 hours before fixation ($n = 3$) and sampled 8-mm wounds every other day for 20 days.

Epidermis

There was a low level of proliferation in the basal cells of the normal epidermis of both *Mus* and *Acomys* of about 6% EDU+ve cells (Figure 2A and B). Following wounding there was a strong proliferative response in the cut epidermis in a halo around the wound such that in both species about 30–40% of the cells have become EDU+ve by day 4 after wounding (Figure 3A). This region of the

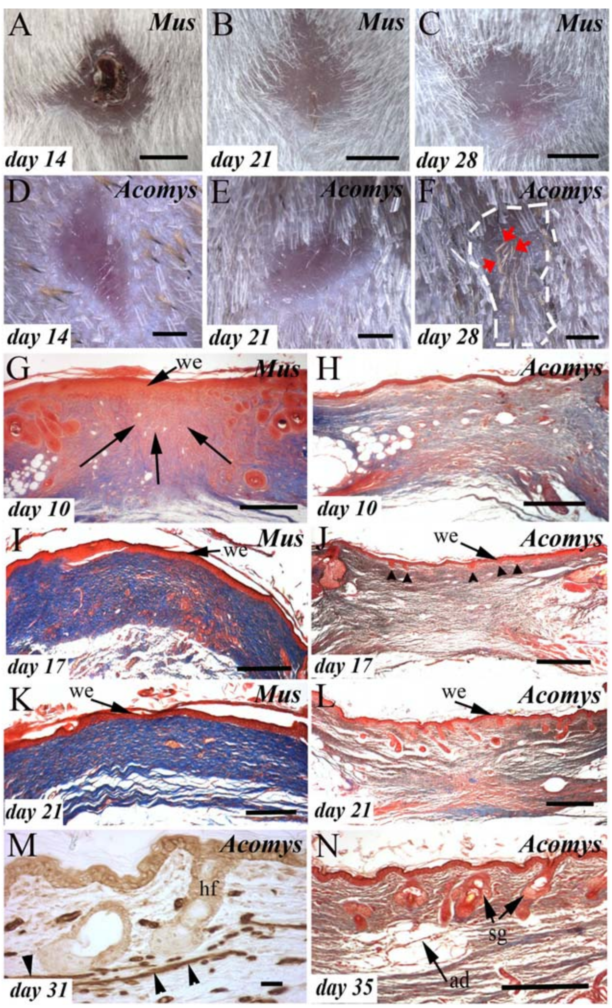

Figure 1. External and histological appearance of *Mus* and *Acomys* 8-mm full thickness skin wounds. (A–C) *Mus* external views at days 14, 21, and 28 showing a clear, hairless epithelium which is considerably smaller than the 8-mm original wound. Bar = 2 mm. (D–F) parallel series of *Acomys* wounds showing the reappearance of golden hairs in the center of the wound at 28 days (red arrowheads). The outer limits of the wound after contraction is marked with a white dotted line. Bar = 2 mm. (G–N) Masson's trichrome stained sections (except M) of wounds, days after wounding marked on each panel. (G) in *Mus* the 10 day granulation tissue is full of densely packed cells which with the matrix gives the appearance of vertical lines of cells. Bar = 500 μm. (H) the same 10 day wound in *Acomys* is filled with a less dense matrix and fewer cells arranged in loose horizontal layers. Bar = 500 μm. I, a dark blue matrix is present under the *Mus* wound epithelium (we) on day 17 which has replaced the predominantly red matrix (in G) with remnants of the initial granulation tissue still present. Bar = 500 μm. (J) in contrast the *Acomys* wound bed hardly stains at all with trichrome revealing a completely different matrix composition. Early hair follicles are present invading the mesenchyme for the wound epithelium (black arrowheads). Bar = 500 μm. (K) by day 21 the *Mus* wound bed is becoming more uniformly blue staining and the wound epithelium (we) is uniform. Bar = 500 μm. (L) the *Acomys* wound bed at day 21 shows hair follicles deeply penetrating the mesenchyme, the layering of the regenerated tissues in the wound bed and only some collagenous matrix in the lowest layer. Bar = 500 μm. (M) immunostaining for SMA reveals strips of smooth muscle (black arrowheads) attaching to the new hair follicles as the erector pili muscles regenerate on day 31. Bar = 100 μm. (N) after 5 weeks adipose tissue begins to reappear at the base of the newly formed hairs (ad). The new sebaceous glands (sg) are now enlarging. Bar = 500 μm.

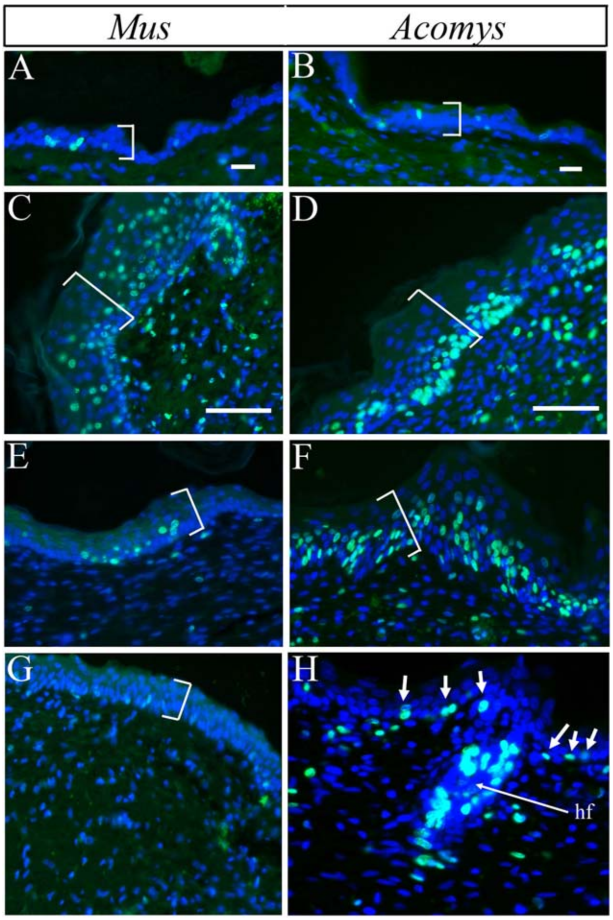

Figure 2. Proliferation in the epithelium before and after wounding in *Mus* (left column) and *Acomys* (right column) measured by EDU uptake. Brackets mark the epithelium. Bars = 100μm. (A) *Mus* normal skin showing few labeled cells. (B) *Acomys* normal skin. (C) epithelium adjacent to the *Mus* wound edge on day 6 showing a large increase in EDU+ve cell numbers (deleted text). (D) wound edge epithelium in day 6 *Acomys* showing a similar result to C. (E) the center of the *Mus* wound epithelium on day 12 showing only a few labeled cells present. (F) by contrast proliferation in the day 12 *Acomys* wound epithelium remains high. (G) the mature wound epithelium on a day 20 *Mus* wound shows low proliferation levels resembling normal skin. (H) in contrast the *Acomys* wound epithelium on day 20 remains proliferatively active because it is generating new hair follicles which have a very high number of EDU+ve cells within them (hf = hair follicle) while the interfollicular epithelium stays above basal levels (white arrows).

epidermis became thickened and labeled cells appeared in the upper layers of the epidermis not just in the basal cells (Figure 2C, D). The rate of proliferation remained high at 30–40% over days 10–12 days of wound healing in this region of epidermis adjacent to the cut edge (Figure 3A). The migrating epidermis, conversely, which migrated over the mesodermal tissues and entered the scab during the first few days after wounding was initially unlabeled suggesting migration and proliferation are incompatible behaviors, but as wound healing progressed proliferation increased to levels compatible with the halo. After day 14 proliferation in the *Mus* wound epidermis dropped precipitously to normal basal levels (Figures 2E and 3A) whereas in *Acomys* proliferation was maintained at levels of 20–30% EDU+ve cells in the wound epithelium between the newly developing hair follicles (Figures 2F and 3A). Proliferation increased even higher in these developing follicles in the *Acomys* wound epithelium, up to levels of 60% and higher (Figures 2H and 3A day 16 follicle, day18 follicle and day 20 follicle) while remaining elevated in the interfollicular wound epithelium (Figures 2H and 3A day 16, day 18 and day 20).

Mesoderm

The initial pattern of mesodermal proliferation in the wounds was very similar in the two species (except for day 4 which showed the only significant difference in the first 2 weeks) and levels increased from 3–4% in normal dermis to a peak of 15–20% +ve cells over the first 2 weeks after wounding (Figure 3B). After this period, in the third week, proliferation in *Mus* mesoderm dropped to the basal rate as the wound deposited collagen and began to develop into an inert scar (Figure 1I and K). In the *Acomys* mesoderm, by contrast, proliferation in the third week dropped from peak levels but remained far above basal levels at 10–15% (Figure 3B) as the tissues of the wound bed began to be regenerated in a scarless manner (Figure 1J and L).

Wound bed matrix and its collagenous composition

In *Mus*, as fibroblasts from the connective tissue layers below the skeletal muscle of the panniculus carnosus invaded the region of the wound soon after damage they deposited a blue collagenous trichrome matrix as they abutted the scab (Figure 4A). As these fibroblasts and other cells displaced the scab they created a wound bed consisting of a dense mass of randomly orientated cells surrounded by a collagenous trichrome matrix (Figure 4C, D). In *Acomys* conversely a similar invasion of cells occurred but without any apparent deposition of a collagenous matrix as the fibroblasts abutted the scab (Figure 4B). The resulting wound bed in *Acomys* was far less dense with cells arranged more loosely and apparently along horizontal strands of matrix (Figure 4E, F) showing completely different trichrome staining characteristics. Cell counts showed a 2-3x higher density of cells in the wound bed in *Mus* than in *Acomys* (Figure 3C). As mesodermal proliferation in the two wounds was very similar (Figure 3B) we assume that there is a greater influx of cells in *Mus* to account for this increased cell density.

This difference in trichrome staining between the two wound beds (Figure 1I–L, Figure 4C–F) suggested that there was likely to be differences in collagen composition in the two types of wound. To examine this we performed microarrays of day 0, day 7, and day 14 samples of *Mus* and *Acomys* wounds and specifically interrogated the data for collagen expression. There were a total of 18 collagen gene isoforms with significantly altered expression levels in *Mus* and *Acomys* compared with normal

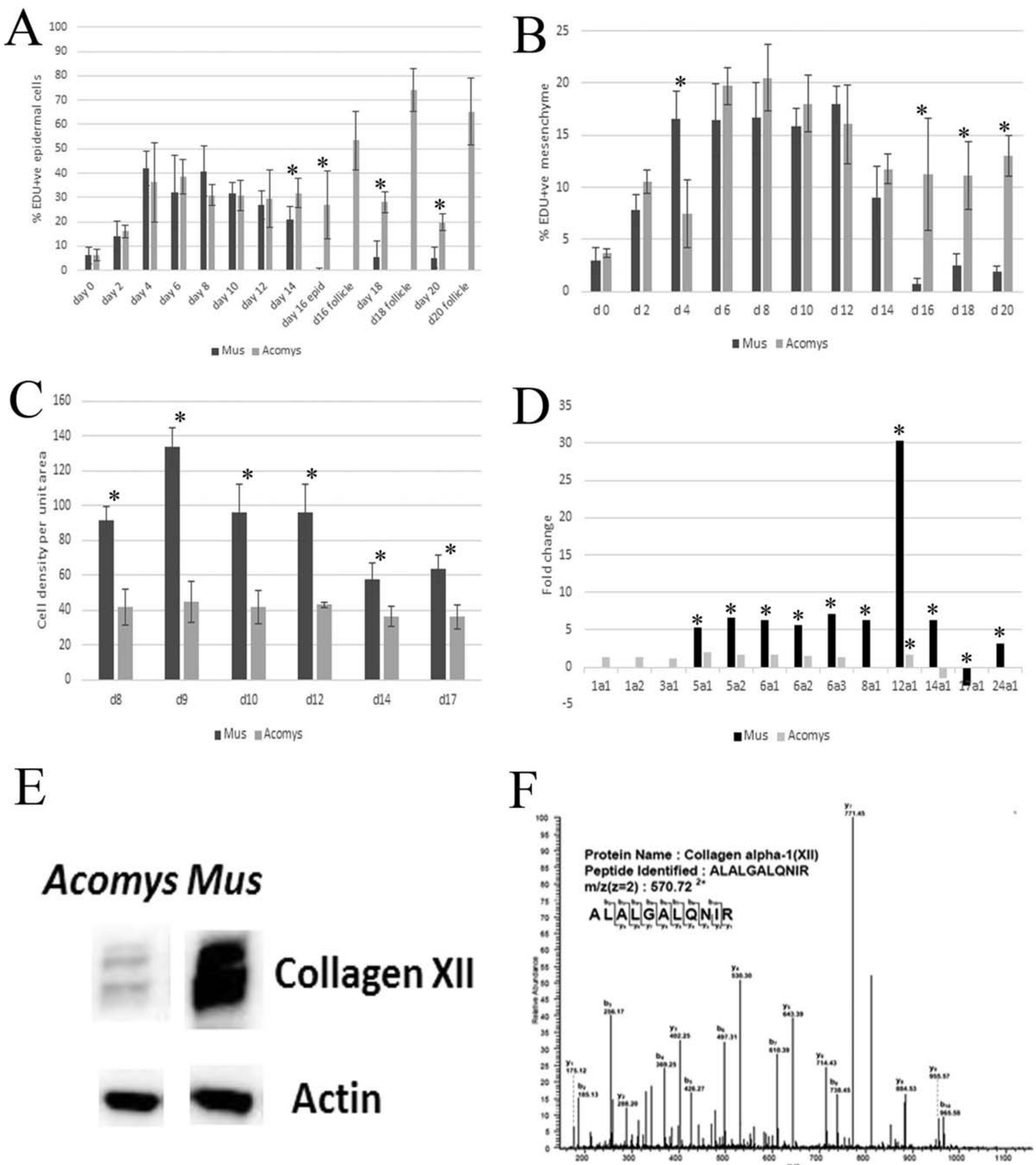

Figure 3. (A) percent EDU+ve epidermal cells in the wound of *Mus* (dark bars) and *Acomys* (light bars) over the first 3 weeks of skin regeneration as assayed by cell counts on sections. At later time periods the *Acomys* wound epidermis counts were separated into new follicles and interfollicular counts. Each column represents 6–12 counts each of 500–1,000 cells. Standard deviations marked and $* = p < 0.01$. (B) percent EDU+ve mesenchymal cells in the wound of *Mus* (dark bars) and *Acomys* (light bars) over the first 3 weeks of skin regeneration as assayed by 6–12 counts each of 500–1,000 cells on sections. Standard deviations marked and $* = p < 0.01$. (C) cell density counts on hematoxylin stained sections of the wound bed (as in Figure 4C and E) in *Mus* (dark bars) and *Acomys* (light bars) on days 6–17 after wounding. Standard deviations marked and $* = p < 0.01$. (D) microarray expression analyses of the collagens expressed in *Mus* (dark bars) and *Acomys* (light bars) day 14 wounds. *Y*-axis shows expression fold-levels relative to day 0 normal skin. *marks significant differences between day 0 and day 14 values for each column. (E) Western blot of day 14 *Acomys* (left lane) and *Mus* (right lane) wounds probed with a collagen 12 antibody showing high levels of this protein in *Mus* and actin as a loading control. (F) identification of peptides by MS: mass spectra of collagen 12.

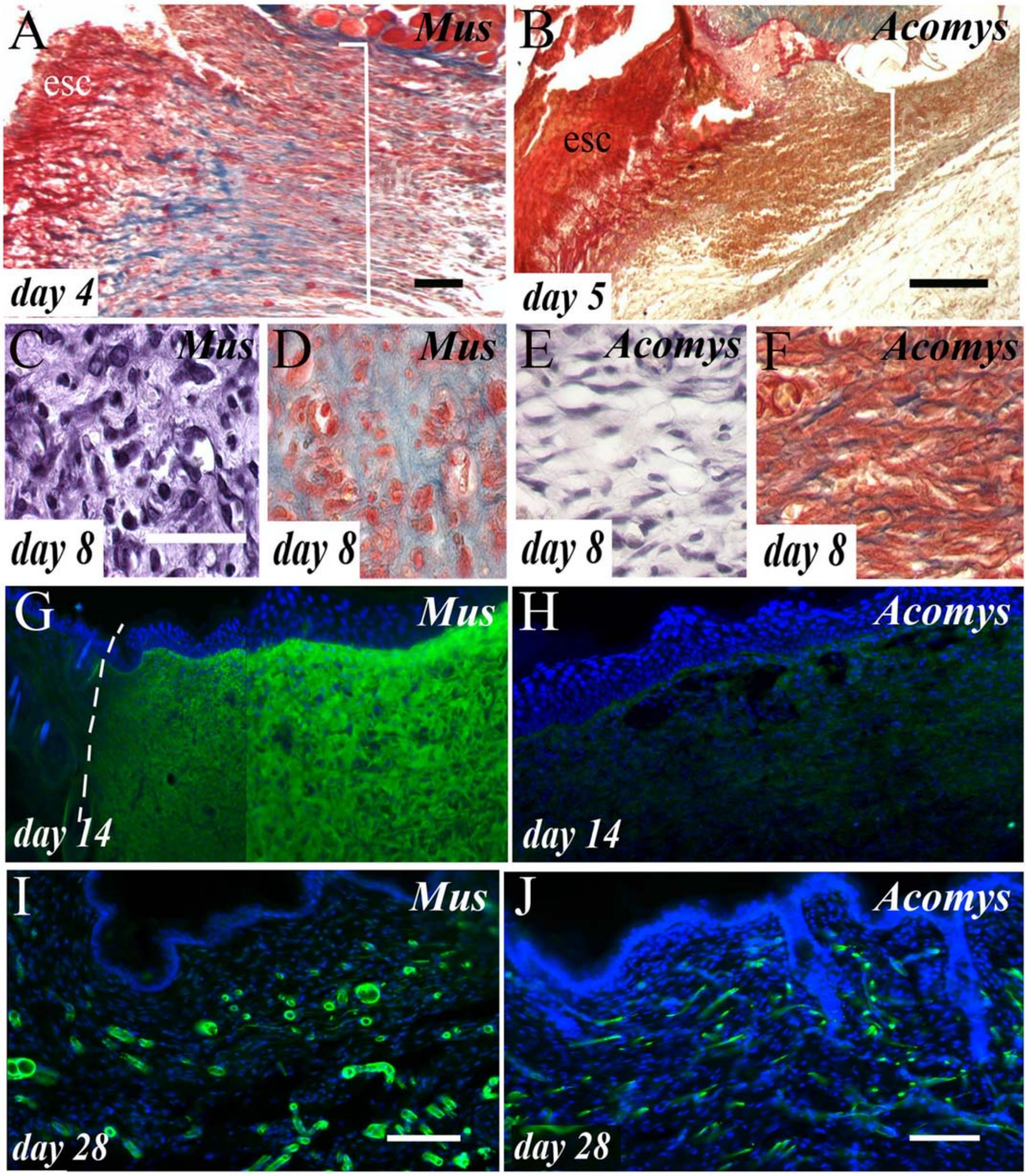

Figure 4. (A) trichrome stained section of the day 4 *Mus* wound showing the region where the cells entering the wound bed (white bracket) meet the eschar (esc). In this region blue, collagenous matrix begins to be deposited. Bar = 100 μm. (B) a similar section to A, but in *Acomys* showing the same cells entering the wound (white bracket) meeting the eschar (esc) but without any blue collagenous matrix deposition. Bar = 100 μm. (C) close-up of a hematoxylin-stained section of the *Mus* wound bed showing a random, swirling arrangement of cells. Bar = 50 μm. (D) trichrome-stained *Mus* wound bed showing blue matrix surrounding the cells. (E) hematoxylin -stained wound bed of *Acomys* show fewer cells than in C and their looser packed, layered arrangement. (F) trichrome-stained *Acomys* wound bed showing very little blue matrix staining. (G–H) collagen XII immunostaining of the day 14 wound bed. (G) *Mus* wound showing intense immunoreactivity in the wound bed with no obvious staining in the normal skin on the edge of the wound (left of the dotted line). (H) same region of an *Acomys* wound bed showing no immunoreactivity. (I–J) SMA staining of the day 28 wound bed to reveal new capillaries. (H) *Mus* wound. Bar = 100 μm. (I) *Acomys* wound revealing smaller, but similar numbers of capillaries (see Figure 6E).

levels in the skin. Concentrating on just the day 14 data set to compare with the histological figures, 13 collagen isoforms showed significantly altered expression levels (Figure 3D). Surprisingly, collagen 1 genes showed no change in *Mus* and a small change in *Acomys* whereas significant up-regulations (5 to 7-fold) were seen in *Mus* with regard to collagens 5, 6, 8, 14, and 24 (Figure 4D). The most dramatic change seen in the *Mus* wound was for collagen 12a1 which was up-regulated 30-fold compared with normal skin (Figure 3D). This prompted us to investigate collagen 12 further to confirm this microarray data and to this end we examined the expression of various collagens in the wound bed by immunohistochemistry on frozen sections of days 10 to 20 wounds. Collagen 1 showed higher immunofluorescence in the *Acomys* wound bed than *Mus,* confirming the microarray data (Figure 3D) which showed a 2.5-fold higher level in *Acomys*. Little collagen 3 could be detected in either species but there was an intense immunoreactivity to collagen 12 in the *Mus* wound (Figure 4G) and no immunoreactivity at all in the *Acomys* wound bed (Figure 4H). Figure 4G shows the edge of the *Mus* wound where there was no immunoreactivity in the normal skin except for part of the hair follicle but as the wound appeared (dotted line marks the edge of the wound) expression levels of collagen 12 rose dramatically. To confirm this difference between the wounds of the two species we also performed Western blots using protein preparations from day 14 wounds and this revealed a greater than 10-fold difference between the expression levels of collagen 12 (Figure 3E).

To confirm that this antibody could identify the *Acomys* homologue of *Mus* collagen 12 we cloned a fragment of the *Acomys* collagen *12a1* gene and determined that its identity to the mouse gene was 96%. The homology of the human sequence to which the collagen 12 antibody was made was 93% and so its absence in the *Acomys* wound bed compared with *Mus* was unlikely to be due to lack of immunoreactivity of the antibody in a different species. Furthermore the antibody reacts with human, mouse, and rat and the homology between *Acomys* and mouse is closer than the homology between mouse and human providing more evidence that the antibody should recognize *Acomys* collagen 12.

We have also conducted a proteomic study of the components of the *Mus* and *Acomys* wound bed which has permitted us to unambiguously identify collagen 12 and to determine its relative levels over time in wounds. The identity of collagen 12 from *Acomys* and *Mus* was determined in tryptic digested samples followed by LC/MS/MS analysis. An example peptide is shown in Figure 3F. Ninety eight common tryptic collagens were identified using a 1% false discovery rate in *Mus* as follows: ALALGALQNIR (aa2387-2397), *p*-value of 6.7×10^{-11}, MW 1139. 38 unique peptides in the N-terminal domain, 46 in the middle fraction and 14 unique peptides in the C-terminal domain were identified. Thus, the sequenced tryptic peptides from *Mus* protein by LC/MS/MS suggested the presence of full length collagen 12 protein.

Collagen 12 quantification was conducted using label free approaches in *Acomys* and *Mus*. The relative amounts of collagen 12 in *Acomys* increased over time with the following changes in amounts: 0.1, 4.7, 5.4, and 13 for 3, 5, 7, and 14 days, respectively. *Mus* showed far higher increases over time with changes of 2.9, 17, 87, and 295 over the same sampling period. The changes were calculated based on log 2 and suggest a 22-fold increase in collagen 12 levels in *Mus* relative to *Acomys* at 14 days.

Immune differences: normal blood cell counts

As one of the major differences between embryonic (scarless) and adult (scarring) wound healing is the composition of the immune system we next asked whether there are any obvious differences in the immune system of *Mus* (scarring) and *Acomys* (scarless) adults by examining white blood cell counts in Geimsa-stained blood smears. In *Mus*, 56% of the leucocytes were identified as lymphocytes, 32% were neutrophils, 6% were monocytes, 5% were eosinophils, and 0.2% were basophils. In *Acomys* the corresponding figures were 79, 6.7, 9.7, 4, and 0.6% (Figure 5A). There was, therefore, a vast decrease in the number of neutrophils in *Acomys* blood (6.7%) compared with *Mus* (32%) which is compensated for by an increase in the number of lymphocytes. *Acomys* is thus very neutropenic.

Immune differences: mast cells and macrophages

In view of the differences in normal leucocyte counts in between *Mus* and *Acomys* we asked whether there were differences in other immune cell types either in normal skin or the wounds. First, examining mast cells using both trichrome blue staining and a tryptase antibody revealed that in the normal skin of *Acomys* (Figure 5E) there were 3.5-fold more mast cells per unit area than in *Mus* skin (Figure 5B and C). In both cases they were most prominent in the papillary dermis around hair follicles. Following wounding, mast cells could be readily detected throughout the *Acomys* granulation tissue (Figure 5F) but were present only at low numbers in the *Mus* wound (Figure 5D) and the same proportional counts were obtained (3.5-fold more in *Acomys*). From this we conclude that the *Acomys* wound contains significantly higher numbers of mast cells than the *Mus* wound and that following wounding mast cells migrate into the granulation tissue from the adjacent dermis in the same proportions.

We next examined the presence of macrophages in wounds using the F4/80 antibody, a widely used marker for mature macrophages. F4/80 macrophages were present in high numbers throughout the *Mus* skin in a halo around the wound, in the scab, and from the beginning of the formation of granulation tissue (Figure 5G). These macrophages remained in the *Mus* wound at all stages, even in 4-week wounds and later (Figure 5H) which had formed a collagenous scar and were going through the slow remodeling phase. In striking contrast the *Acomys* wounds never contained any F4/80 macrophages at any stage (Figure 5I) even though they could be seen in the normal skin around the wound and in the connective tissue which had been damaged below the panniculus carnosus muscle. F4/80 macrophages were also detected in the spleen of *Acomys* (Figure 5J) confirming that it was not the inability of this mouse antibody to react with a different species that caused this absence of immunoreactivity in the *Acomys* wound bed.

To further explore this lack of macrophages in the *Acomys* wound bed we examined early wounds on day 3 and 5 for markers of classical and alternative macrophage activation by RT- PCR. In Mus, *iNos* (classical activation) was strongly up-regulated on both days relative to normal skin, as was *Dectin-1* and *Arg-1* (alternative activation) (Figure 5K), but there was *no expression* of these genes detectable in the *Acomys* wound. To demonstrate that this was not due to the failure of the primers to anneal to the relevant *Acomys* transcripts we cultured *Acomys* spleen cells and activated them with pokeweed mitogen. Under these conditions *Ifn-γ* and *iNos* (classical activation) and *Dectin-1* and *Mro-1* (alternative activation) were readily detectable in *Acomys* spleen cells (Figure 5L). From this, we conclude that there are no mature macrophages in the *Acomys* wound bed and this is not due to our inability to detect them.

Other macrophage markers are, however, detectable with similar distributions over time in *Mus* and *Acomys*

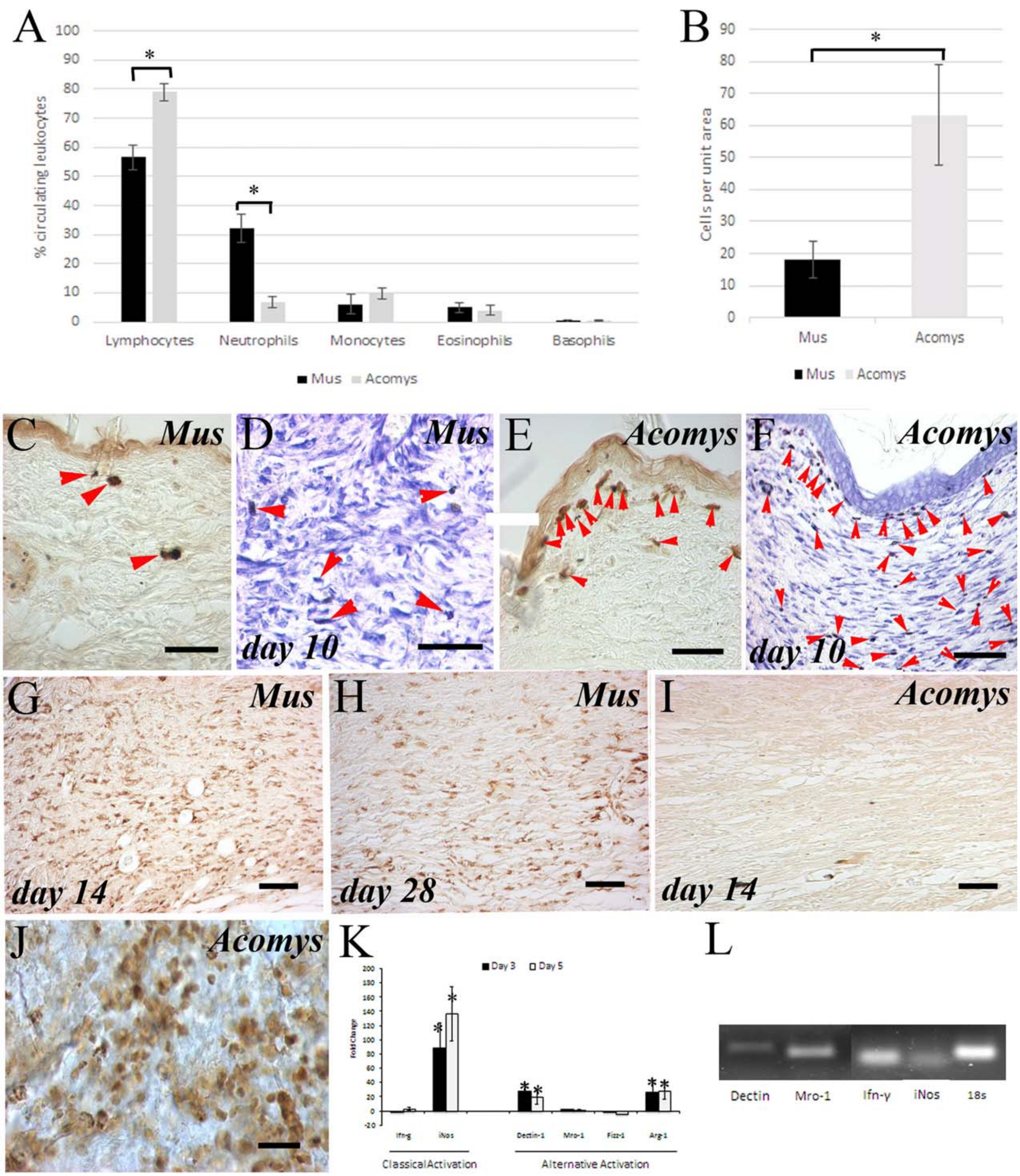

Figure 5.

wounds. Using MOMA-2, a marker for monocytes/macrophages we observed that both wounds were devoid of MOMA-2 immunoreactive cells until the middle of the third week when the wound bed was in the remodeling phase (day 16 onward, data not shown). The cells that were identified by this antibody were of a totally different morphology from those reacting with F4/80 being large and round cells rather than thin and elongated and added further weight to the conclusion that a specific set of macrophages, the F4/80 cells, were absent in *Acomys* skin wounds.

Cytokines present in the wounds

As macrophages are a major source of chemokines, cytokines, and growth factors in the wound and respond themselves to chemoattractants we also asked whether differences in the cytokine content of *Mus* and *Acomys* wounds could be detected. To undertake this we prepared protein extracts of normal skin and wounds at 0, 3, 5, 7, and 14 days ($n = 3$) and used a mouse cytokine array kit (R & D Systems) to simultaneously detect 40 cytokines. Even though we were using *Mus* antibodies on these arrays to detect *Acomys* proteins, all our immunocytochemical analyses have successfully used such antibodies and we have not failed to detect the relevant proteins in any *Acomys* tissue suggesting that the polyclonal antibodies that we have used cross-react successfully with this species as well as mouse, rat, or human. Furthermore, our analyses of *Acomys* proteins such as collagen 12 described above which have been cloned and sequenced have revealed an extremely high degree of homology (>90%) between *Acomys* and *Mus*.

Using these arrays we can detect 30 out of the 40 cytokines present on the array in *Mus* wounds. They show a strikingly consistent time course being present at barely detectable levels in normal skin then rising to a peak at day 5 and tailing off on day 7 to become undetectable again by day 14 (Figure 6A–C, black columns). Their relative levels varied and the maximum levels were detected for G-CSF, CXCL1, MIP-1β, C5/C5a, MCP-1/CCL2, MIP-1α, MIP-2 (Table 1).

In *Acomys* wounds there were dramatic differences as 18 of these cytokines present in *Mus* were undetectable in *Acomys* (Table 1). These included cytokines secreted by macrophages (GM-CSF, G-CSF, CXCL10, CXCL11, MIP-1β) e.g., Figure 6A and macrophage chemoattractants/inducers (CCL1, MCP-5, M-CSF) whose absence correlated with the lack of F4/80+ve macrophages in the *Acomys* wounds. One of the cytokines most significantly absent from the point of view of the difference in composition of the matrix of the wounds in these two species (Figure 1) is the lack of TIMP-1 in *Acomys* (Table 1, Figure 6B). As these molecules inhibit the action of MMPs this would lead to a less rigid structural integrity of the *Acomys* wound. There were six cytokines present in the *Acomys* wound at lower levels than in *Mus* (C5/C5a, IL-16, MCP-1. MIP-1α, MIP-2, TNF-α) and five cytokines present at the same levels as *Mus* (CXCL13, IFN-γ, IL-1α, IL-1β, IL-1ra) e.g., Figure 6C. Only 1 cytokine was present at higher levels in *Acomys* than in *Mus* and this was RANTES (Figure 6D).

Vascularity of wounds

In view of the lack of mature macrophages in the wound bed of *Acomys* and their role in promoting vascularity of the granulation tissue[31] we finally considered whether there was a difference in the degree of vascularity in *Mus* and *Acomys* wounds. We investigated this aspect of *Mus* and *Acomys* wounds by staining sections of the wound bed over a 4-week time course with a SMA antibody to identify the invading capillaries and quantified their number. In both species the granulation tissue of the wound bed did not become revascularized with recognizably organized capillaries until the beginning of the second week after wounding. After this time capillary ingrowth could readily be seen using a SMA antibody (e.g., Figure 4I, J). Initially in *Mus* the invading capillaries took an upward sweeping course resembling the matrix structure of the Mus wound bed (Figure 1G) whereas in *Acomys* the capillaries were more randomly oriented. Counts of the numbers of SMA+ve capillaries per unit area showed no significant differences between the two (Figure 6E) except for one sample time, day 20. We also examined

Figure 5. (A) White blood cell counts in *Mus* (dark bars) and *Acomys* (light bars) blood smears. Neutrophil counts are down from 32% in *Mus* to only 7% in *Acomys* and the difference is made up by an increased number of lymphocytes. Standard deviations marked and $* = p < 0.001$. (B) counts of mast cells per unit area in normal skin of *Mus* (dark bar) and *Acomys* (light bar). Standard deviations marked and $* = p < 0.001$. Comparable histological sections are shown in C and E. (C) tryptase antibody staining in normal *Mus* skin reveals three mast cells (red arrowheads) in this area of the dermis. Bar = 100 μm. (D) toluidine blue staining of the day 10 *Mus* wound bed just below the wound epithelium reveals some mast cells (red arrowheads). Bar =100 μm. (E) an equivalent area to A in *Acomys* normal skin reveals many more mast cells by tryptase antibody staining (red arrowheads). Bar =100 μm. (F) an equivalent area to B in the *Acomys* wound reveals far more toluidine blue stained mast cells (red arrowheads). Bar =100 μm. (G–J) F4/80 immunostaining to detect the presence of macrophages. Bars =100 μm. (G) day 14 *Mus* wound bed showing many F4/80+ve cells throughout all levels of the resolving granulation tissue. Bar = 100 μm. (H) the day 28 *Mus* wound reveals the continuing presence of macrophages throughout the wound bed. Bar = 100 μm. (I) an equivalent region in the *Acomys* day 14 wound bed to G reveals the complete absence of F4/80+ve cells. Bar = 100 μm. (J) F4/80 staining of the *Acomys* spleen reveals many positive macrophages. (K) RT-PCR of *Mus* wounds on day 3 (black bars) and day 5 (white bars) showing detectable up-regulation of three genes which are known to be expressed in *Mus* macrophages after cytokine activation. *marks statistical significance compared with day 0 value. (L) polymerase chain reaction (PCR) gel of mitogen activated *Acomys* spleen cells showing detectable levels of four genes known to be involved in macrophage activation.

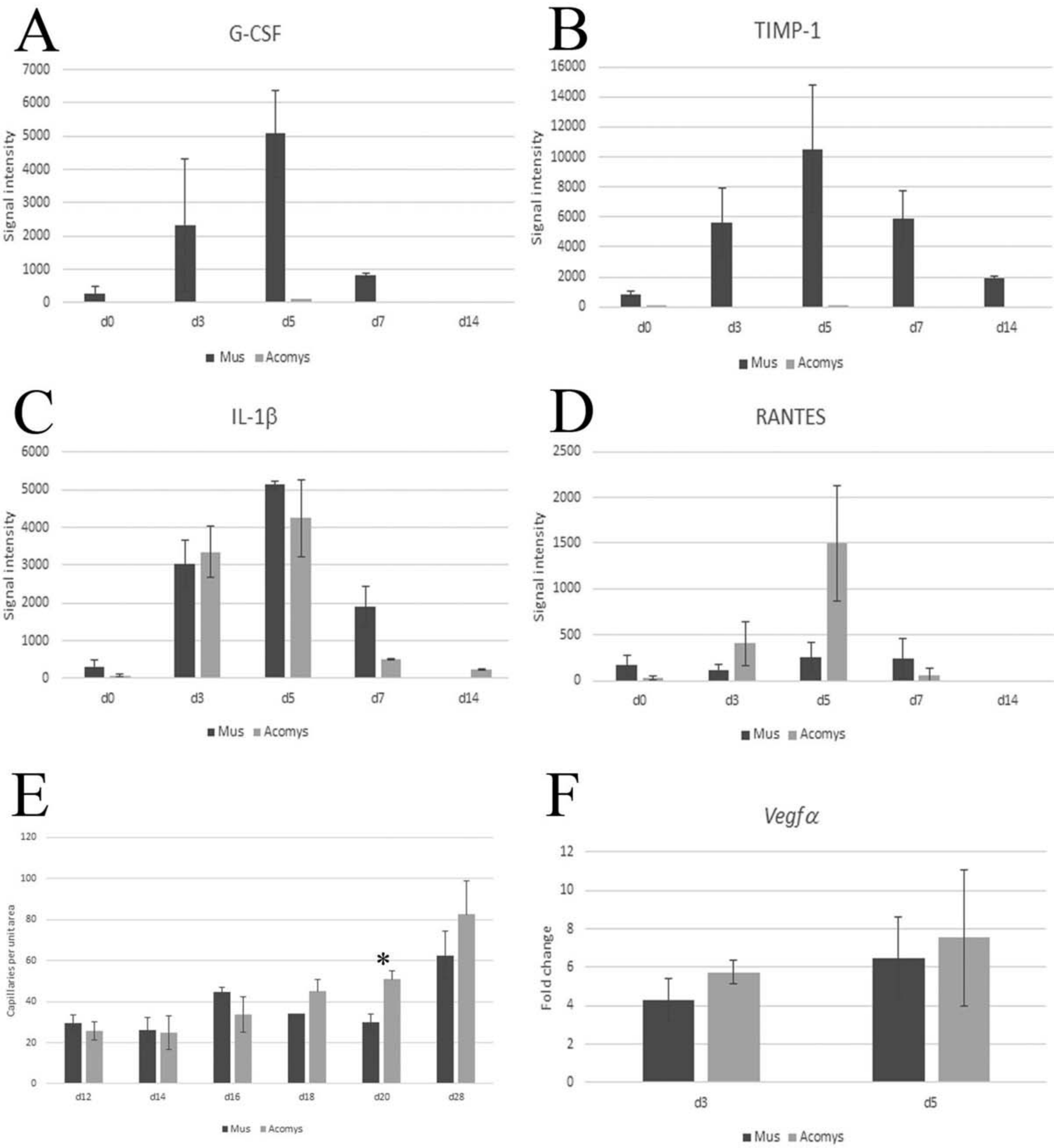

Figure 6. (A–D) levels of four cytokines in the wound at day 0 (unwounded skin), 3, 5, 7, and 14 in *Mus* (dark bars) and *Acomys* (light bars). Standard deviations are marked. A, G-CSF levels showing a typical *Mus* response peaking at day 5 and an absence in *Acomys*. (B) TIMP-1 levels showing the typical response in *Mus* and an absence in *Acomys*. (C) IL-1β an example of comparable levels between *Mus* and *Acomys*. (D) RANTES is the only cytokine which shows higher levels in *Acomys* than *Mus*. (E) counts of number of SMA+ve capillaries per unit area on sections of *Mus* (black bars) and *Acomys* (gray bars) wounds over days 12–28 after wounding. Only the day 20 data point shows a significant difference between *Mus* and *Acomys* ($* = p < 0.01$). Standard deviations are marked. (F) RT-PCR showing comparable *Vefg*α levels in day 3 and day 5 wounds of *Mus* (black bars) and *Acomys* (gray bars) wounds.

the expression of *Vegf*α by RT-PCR which revealed no significant differences in the levels of this gene on day 3 or day 5 after wounding (Figure 6F). Later, however, microarrays revealed that on day 14 this gene showed a 3.1-fold up-regulation in *Acomys* compared with *Mus* (data not shown).

DISCUSSION

We describe here some of the cellular and molecular differences between the response of *Acomys* and *Mus* skin to full thickness wounding which could potentially be responsible for the dramatic difference in the response of these

Table 1. Summary of the relative maximum levels of cytokines detected in *Mus* and *Acomys* wounds over days 3–14. Maximum levels = **+++++**, minimum levels = **+**, levels in between represented by differing numbers of plusses, absence = **–**

Cytokine	Mus	Acomys
GM-CSF	++	–
G-CSF	+++++	–
CCL1	+	–
sICAM-1/CD54	++++	–
IL-2	+++	–
IL-4	+	–
IL-6	++++	–
IL-7	+	–
IL-13	+	–
CXCL10	++	–
CXCL11	+	–
CXCL1	+++++	–
M-CSF	+++	–
MCP-5/CCL12	++	–
MIG/CXCL9	+	–
MIP-1β	+++++	–
TREM-1	++++	–
TIMP	++++	–
C5/C5a	+++++	+
IL-16	++	+
MCP-1/CCL2	+++++	+
MIP-1α	+++++	++++
MIP-2	+++++	+
TNF-α	++	+
CXCL13	+	+
IFN-γ	+	+
IL-1α	++	++
IL-1β	++++	++++
IL-1ra	++++	++++
RANTES	+	++

two organisms to skin damage—*Acomys* regenerates hairs, sebaceous glands, smooth muscle, dermis, and skeletal muscle in a scar-free manner whereas *Mus* produces a hairless scar.

Despite being an adult mammal our preliminary analyses of the *Acomys* immune system resembles that of the regenerating lower vertebrates and the mammalian embryo in being "blunted." Urodeles, which can regenerate their skin in a scar-free manner,[3,4] are immunodeficient with minimal inflammatory response and a weak cellular immunity[22] a description which would also apply to the nonscarring mammalian embryo.[8–11] There are several relevant differences in the *Acomys* immune system compared with *Mus*.

First, their blood is neutropenic with a greatly reduced number of neutrophils, as is that of the embryo,[9] supporting the idea that neutrophils contribute to the scarring phenotype. However, reducing or eliminating neutrophils does not seem to lead to any improvement in scarring and repair[32] despite an increased rate of reepithelialization[33] so it is unlikely that the reduced number of neutrophils alone is responsible for the regenerative behavior of *Acomys* skin. Second, another immune cell, the mast cell, shows a difference in the opposite direction. There are 3.5-fold more mast cells both in the normal *Acomys* skin and in the *Acomys* wound which runs counter to the data from scar-free embryonic healing where there are fewer mast cells.[19,20] The administration of mast cell lysates to early embryos induces scarring and smaller scars are generated in late embryos from mast cell deficient mice so this difference does not seem to be responsible for the scar-free behavior of *Acomys* skin.

Third, the most significant immune cell difference between *Acomys* and *Mus* is likely to be the absence of F4/80$^+$ macrophages in the *Acomys* wound bed. This is not because the antibody does not react with *Acomys* macrophages as these cells can readily be seen in the spleen and splenic cells can be activated to express typical markers such as *Dectin*, *Mro*-1, *Ifn-γ* , and *iNos*. This lack of macrophages in the *Acomys* wound bed reflects a fourth major difference in the immune system, the vastly reduced wound cytokine response in *Acomys* reflected in the absence of many cytokines normally present in the wound. Of 30 cytokines identified in the *Mus* wound bed only 12 could be detected in *Acomys* with 6 of these at much lower levels, 5 at similar levels, and only 1, RANTES, being present at increased levels.

The *absence* of many of these pro-inflammatory molecules and/or macrophages is very likely to be the major contributor to the lack of scarring in *Acomys* as improved skin healing in terms of a reduced scar and reduced fibrosis levels in other organs is characteristic of cytokine knockouts and macrophage level manipulations in *Mus*. Macrophages are considered to play an important role in wound healing by phagocytosing neutrophils, antigen-presentation, production of cytokines, and chemokines to stimulate capillary growth, fibroblast proliferation, and collagen synthesis.[34] When macrophages are depleted from the *Mus* wound the effects are usually described as detrimental, but several of the effects are very similar to what we see, here, in the regenerating *Acomys*: reduced granulation tissue formation, reduced collagen deposition and reduced scar size.[31,35] Similarly, in neonatal PU.1 null mice which lacks both macrophages and neutrophils wound healing is scarless.[36] Not all the effects seen in these macrophage depletion experiments were seen in *Acomys*, however, as we detected no delay in reepithelialization, no reduction in angiogenesis and no reduction in fibroblast proliferation which were the same in the two species following skin wounding.

Many of the individual cytokines which could not be detected or were present at very low levels in *Acomys* wounds when attenuated in mice give greatly reduced fibrotic phenotypes not only in the skin but after damage to other organs as well. MCP-1 was present at the highest levels in *Mus* wounds but only at very low levels in *Acomys* wounds (Table 1). MCP-1 deficient mice show reduced numbers of macrophages, reduction of collagen 1 levels, a reduction in fibrosis and minimal scarring after skin damage,[37,38] kidney tubular damage,[39] and during atherosclerosis.[40,41] Another high level cytokine in *Mus* and absent from the *Acomys* wound was IL-6. Removal of this cytokine protects the lung from pulmonary fibrosis.[42] TREM-1, which was absent in *Acomys* wounds, amplifies the production of proinflammatory cytokines, is up-regulated after myocardial infarction and in its absence there is less fibrosis and a smaller infarc size[43] C5 deficient mice have accelerated cutaneous wound healing and a decreased inflammatory infiltrate[44] just as the *Acomys* wound does. Thus, there is considerable evidence to support the conclusion from our work reported here that the decreased levels or absence of many cytokines in *Acomys*, many of which are known to be proinflammatory, is at least in part responsible for the absence of macrophages, a decreased fibrotic response and the lack of scarring.

We also revealed that the decreased fibrotic response seen in *Acomys* was primarily associated with the failure to up-regulate several collagens in the wound bed which along with the absence of TIMP-1 would likely explain the dramatic difference in trichrome staining between the two wounds (Figure 1). The most spectacular difference was in collagen 12 which was up-regulated between 10 and 30-fold in *Mus* in day 14 samples. We used proteomics to unambiguously confirm the microarray and immunofluorescence data that this protein was indeed collagen 12 which was switched on precisely in the *Mus* wound bed. Collagen 12 is a fibril-associated collagen with interrupted triple helices (FACIT) and along with collagen 14 binds the fibrillary collagens such as collagen I to stabilize the ECM.[45] It is present in dense connective tissues such as bone and tendon as well as the skin and cornea[46,47] and it also binds other ECM components such as decorin, tenascin-X, and fibromodulin. Thus, the presence of collagen 12 in the *Mus* wound bed would surely lead to a vastly different matrix composition and it would be of interest to examine the wound healing and scarring properties of the collagen 12 mutant mouse which has so far only been characterized as possessing an abnormal corneal endothelium.[48]

In summary, these observations on the regenerating, scar-free healing of excisional wounds of *Acomys* provide a starting point for revealing causal differences between scarring and nonscarring skin and a test-bed for devising strategies to prevent scarring in humans.

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